

# Structural Aliasing Analysis in Time-Series Forecasting Using Chronos

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## Executive Summary

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## Time-Series Forecasting

Time-series forecasting stands as a cornerstone of modern predictive analytics, vital for interpreting and anticipating the dynamic, state-dependent behaviors inherent to complex systems. Whether predicting fluctuating loads in renewable energy grids, identifying macro-economic trends, or monitoring structural health in industrial machinery, the ability to project historical sequential data into future horizons shifts

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During the preparation of this work, the authors used AI-based tools, including GPT-5.5, Gemini 3.1 Pro, Gemini 3.5 Flash, Sonnet 4.6, and Opus 4.7, to support language refinement and improve clarity. All generated or suggested content was carefully reviewed, revised where necessary, and validated by the authors, who assume full responsibility for the final content of this work.

organizational strategy from reactive mitigation to proactive optimization. As systems grow increasingly interconnected and generate data at unprecedented scales, the demand for forecasting frameworks capable of parsing intricate temporal dynamics has driven a massive evolution in modeling methodologies.

## Signal Representation and Aliasing Vulnerabilities

Regardless of the downstream modeling architecture utilized, the predictive fidelity of any forecasting pipeline remains strictly bounded by the digital representation of the underlying signal. This representation relies heavily on initial data acquisition, signal conditioning, and pre-processing protocols.

The fundamental mathematical boundary governing the digitization of continuous physical processes is the Nyquist-Shannon sampling theorem. The theorem dictates that a continuous-time signal containing frequency components can be perfectly captured and reconstructed without loss of information if and only if it is sampled uniformly at a sampling frequency  $f_s$  that is strictly greater than twice the highest frequency component ( $f_{\max}$ ) present within the signal:

$$f_s > 2f_{\max}. \quad (1)$$

The threshold  $f_N = \frac{f_s}{2}$  is defined as the Nyquist frequency. When a system violates this criterion, either due to an insufficient sampling rate or the absence of a proper low-pass anti-aliasing filter, high-frequency components beyond the Nyquist limit fold back into the lower spectrum. This distortive phenomenon, known as *aliasing*, introduces spurious, artificial low-frequency components that permanently mask and corrupt the true underlying dynamics of the signal.

## Evolution of Forecasting Methodologies

Historically, time-series analysis relied on classical statistical models such as AutoRegressive Integrated Moving Average (ARIMA) and Exponential Smoothing (ETS). These traditional methods are deeply rooted in mathematical theory, offering high interpretability, exact statistical guarantees, and robust performance on stable, low-dimensional datasets. However, they rely heavily on strict underlying assumptions such as linearity, and stationarity.

To capture complex, non-linear temporal behaviors without rigid feature engineering, the paradigm shifted toward deep learning. Early architectures leveraged Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and Convolutional Neural Networks (CNNs) to map temporal sequences sequentially.

More recently, the state of the art has been dominated by transformer-based architectures. By replacing recurrent bottlenecks with self-attention mechanisms, transformers can process sequences in parallel and explicitly compute dependencies between distant time steps. This allows the model to learn both localized high-frequency fluctuations and long-range macro-patterns simultaneously, forming highly adaptive architectures that can be fine-tuned to specific downstream industrial or scientific applications.

Representing the cutting edge of this evolution is the emergence of time-series foundation models, most notably exemplified by the *Chronos* family of frameworks. Unifying time-series forecasting with the architectures powering modern natural language processing, Chronos treats time-series values not as continuous regressions, but as discrete language tokens.

The framework tokenizes real-valued historical data by quantizing values into discrete buckets via a scaling and binning operator. These numerical tokens are then fed directly into standard transformer language model backbones (such as T5). Trained on billions of diverse data points spanning synthetic and real-world domains, Chronos exhibits powerful zero-shot generalization, accurately predicting future trajectories across entirely unseen systems without requiring task-specific structural retraining or structural fine-tuning. This reliance on discrete tokenization windows, however, introduces novel architectural interactions with signal frequencies, exposing the model to localized structural vulnerabilities when processing periodic inputs.

## Patch-Based Tokenization: a Formal Description

Let  $\mathbf{x} = \{x[n]\}_{n \in \mathbb{Z}}$  be a discrete-time signal representing the input time series. We define the patch-based tokenization operator  $\mathcal{T}_{P,S}$  with patch length  $P \in \mathbb{N}^+$  and stride  $S \in \mathbb{N}^+$  as a mapping that transforms the signal  $\mathbf{x}$  into a sequence of vector-valued tokens  $\mathbf{T} = \{\mathbf{t}_k\}_{k \in \mathbb{Z}}$ , where each token  $\mathbf{t}_k \in \mathbb{R}^P$ .

The  $k$ -th token is defined as

$$\mathbf{t}_k = \mathcal{T}_{P,S}(\mathbf{x})[k] = \begin{bmatrix} x[kS] \\ x[kS + 1] \\ \vdots \\ x[kS + P - 1] \end{bmatrix}. \quad (2)$$

Using element-wise indexing notation,

$$\mathbf{t}_k[m] = x[kS + m], \quad m \in \{0, \dots, P - 1\}. \quad (3)$$

### Phase-Locking and Patch Indistinguishability

Consider a continuous-time periodic signal sampled uniformly at sampling frequency  $f_s$ , yielding the discrete-time sequence  $x[n]$ . We isolate a fundamental harmonic component and model it as

$$x[n] = \sin\left(2\pi \frac{f}{f_s} n + \phi\right), \quad (4)$$

where  $f$  denotes the signal frequency satisfying the Nyquist criterion  $f < \frac{f_s}{2}$ , and  $\phi \in [0, 2\pi)$  is an arbitrary phase offset.

Phase-locking occurs when the structural phase profile of the signal repeats exactly under a translational shift equal to the stride  $S$ . Evaluating a stride translation yields

$$x[n + S] = \sin\left(2\pi \frac{f}{f_s} (n + S) + \phi\right) = \sin\left(2\pi \frac{f}{f_s} n + 2\pi \frac{fS}{f_s} + \phi\right). \quad (5)$$

For exact alignment between the signal and the tokenization operator, the phase accumulation over one stride must correspond to an integer number of  $2\pi$ -periods

$$2\pi \frac{fS}{f_s} = 2\pi c, \quad c \in \mathbb{N}^+. \quad (6)$$

As a result, the signal exhibits exact stride-period translational invariance

$$x[n + S] = \sin\left(2\pi \frac{f}{f_s} n + 2\pi c + \phi\right) = \sin\left(2\pi \frac{f}{f_s} n + \phi\right) = x[n]. \quad (7)$$

Solving Equation (6) for  $f$  yields the phase-locking frequency class, which is the class of frequencies that produce exact stride-period translational invariance, we denote that as  $\mathcal{F}_{\text{lock}}$ :

$$\mathcal{F}_{\text{lock}} = \left\{ f \in \mathbb{R}^+ \mid f = c \frac{f_s}{S}, \ c \in \mathbb{N}^+, \ f < \frac{f_s}{2} \right\}. \quad (8)$$

The collapse of consecutive tokens is not restricted to sinusoidal signals. Consider an arbitrary discrete-time signal satisfying the translational invariance condition

$$x[n + S] = x[n], \quad \forall n. \quad (9)$$

Taking into account two consecutive tokens, using the element-wise notation from Equation (3) and applying Equation (9) gives

$$\mathbf{t}_{k+1}[m] = x[(k+1)S + m] = x[kS + m + S] = x[kS + m] = \mathbf{t}_k[m]. \quad (10)$$

Since this equality holds for every component,

$$\mathbf{t}_{k+1} = \mathbf{t}_k. \quad (11)$$

Consequently, any signal exhibiting exact stride-period translational invariance produces identical consecutive tokens regardless of the patch length  $P$ .

## Characterization of the Loss of Observability

When consecutive patches become indistinguishable, a localized structural aliasing event occurs. This degradation can be characterized through three complementary perspectives.

### 1. Rank Deficiency in the Embedding Space

Let

$$\mathbf{M} = [\mathbf{t}_k, \mathbf{t}_{k+1}, \dots, \mathbf{t}_{k+L-1}] \in \mathbb{R}^{P \times L} \quad (12)$$

be the local trajectory matrix.

If two consecutive tokens become identical, then the corresponding columns become linearly dependent. Consequently, the trajectory matrix cannot achieve full rank:

$$\text{rank}(\mathbf{M}) < \min(P, L). \quad (13)$$

More generally, every duplicated token introduces at least one linear dependency and therefore cannot increase the matrix rank. Repeated patch collapse systematically compresses the effective dimensionality of the local trajectory manifold and reduces the geometric diversity available to downstream embedding layers.

### 2. Content Degeneration in Self-Attention

If consecutive patches satisfy

$$\mathbf{t}_k = \mathbf{t}_{k+1}, \quad (14)$$

their content representations become identical.

For transformer architectures without positional information,

$$\mathbf{q}_k = \mathbf{q}_{k+1}, \quad \mathbf{k}_k = \mathbf{k}_{k+1}. \quad (15)$$

Consequently, rows  $k$  and  $k+1$  of the content-derived attention score matrix become identical.

Modern transformer architectures generally employ positional encodings, rotary embeddings, or relative positional biases. Such mechanisms can still distinguish consecutive positions. Nevertheless, the content-derived component of the attention mechanism becomes degenerate, forcing the model to rely increasingly on positional information to break the symmetry introduced by token collapse.

### 3. Loss of Sub-Stride Observability

The tokenization operator observes the signal only at stride-separated locations. This induces an effective observation frequency

$$f_{\text{eff}} = \frac{f_s}{S}, \quad (16)$$

Signal dynamics occurring at temporal scales shorter than the stride interval become progressively harder to observe through the token sequence.

Under phase-locking conditions, consecutive tokens become identical and the local representation collapses into a stationary state. Fine-grained phase evolution and high-frequency variations occurring between token boundaries therefore become hidden from the observable representation presented to the model.

## Analysis of Structural Degeneracies and Invariances

When  $f \in \mathcal{F}_{\text{lock}}$ , the resulting behavior depends on the relationship between  $P$  and  $S$ .

**Case 1: Equal Patch Size and Stride ( $P = S$ )** When patches are contiguous and non-overlapping,

$$P = S. \quad (17)$$

Equation (9) directly implies

$$\mathbf{t}_{j+1} = \mathbf{t}_j \quad \forall j. \quad (18)$$

The token sequence collapses into a static series of identical vectors, eliminating all observable temporal variation.

**Case 2: Overlapping Windows ( $S < P$ )** When the stride is smaller than the patch length,

$$S < P, \quad (19)$$

the translational invariance condition remains valid:

$$x[n + S] = x[n]. \quad (20)$$

Therefore,

$$\mathbf{t}_{j+1} = \mathbf{t}_j. \quad (21)$$

Even though each token spans a larger temporal context, exact stride-period alignment forces consecutive tokens to remain identical. Consequently, overlap alone cannot prevent token-space degeneracy.

**Case 3: Disjoint Windows ( $S > P$ )** When

$$S > P, \quad (22)$$

gaps of length  $S - P$  appear between successive windows. Under the phase-locking condition, which still guarantees

$$\mathbf{t}_{j+1} = \mathbf{t}_j. \quad (23)$$

However, the interval separating consecutive windows remains completely unobserved by the tokenization operator. Consequently, distinct signals may produce exactly the same token sequence despite exhibiting different dynamics within the hidden intervals.

In this regime, the observation operator becomes non-injective: multiple underlying trajectories can map to an identical token representation. This creates a stronger form of information loss than simple token degeneracy because portions of the signal become permanently unrecoverable from the tokenized representation.

## Domain Description and Ontology Representation

### Domain Description: Cyber-Physical PV Micro-Grid with Structural Aliasing Awareness

The considered domain takes into account the energy generated by an asset infrastructure centered around a **PhotovoltaicArray**, operating within a localized smart micro-grid ecosystem. This environment is dynamically monitored, analyzed, and managed by an autonomous AI **ControlAgent** that balances power flows, detects system failures, and evaluates data integrity. Uniquely, the domain bridges physical power generation with data science tokenization constraints, mapping how time-series windowing chunking strategies (**Chronos patch tokenization**) impact the data fidelity fed into machine learning models.

#### Physical Assets and Grid Topology

The physical core of the system is composed of four primary entities under the **PhysicalComponent** class, which are physically **connectedTo** one another to form the micro-grid’s electrical topology:

- The **PhotovoltaicArray** serves as the primary generation source, defined by its physical size (**panelCount**) and inverter nameplate capacity (**nominalPowerW**).
- The array directly **suppliesTo** three downstream assets: the **NationalGrid** interface (an infinite bidirectional bus for energy exchange), local **InternalLoad** units (representing dynamic consumption demand), and a **BatteryStorage** unit.
- Functionally, the **BatteryStorage** unit actively **buffers** the **PhotovoltaicArray**, absorbing excess generation or discharging power during deficits.

#### Telemetry Streams and Tokenization Layer

To evaluate system performance, data streams are continuously collected across two operational vectors:

- The **PhotovoltaicArray** **measures** a continuous **Signal**, which represents the raw time-series telemetry stream. This stream encapsulates instantaneous data properties such as photovoltaic power (**pvPowerW**), global horizontal irradiance (**horizontalIrradianceWm2**), and tilted surface irradiance (**tiltedIrradianceWm2**), alongside frequency-domain metadata including the raw sampling frequency, fundamental wave frequency, amplitude, and phase.
- Concurrently, the array is mapped via a **hasWeatherObservation** relationship to a **WeatherObservation** snapshot, which logs environmental metrics like ambient air temperature (**airTemperatureC**) and wind vectors.

To prepare this continuous telemetry for deep learning inference, the raw **Signal** passes through a processing layer that **tokenizes** it into a **PatchedSignal**. This represents the sequence chunking strategy employed by foundation architectures like Chronos, configured via discrete attributes: **patchSize** (samples per token), **stride** (step size between windows), and an **overlapRatio**.

#### Inferred Environmental Contexts and System Anomalies

An execution reasoner dynamically updates the operational metadata of the system based on physical and mathematical rules. The system extracts a baseline **GenerationState** derived entirely from the tilted solar irradiance ( $G_{\text{tilt}}$ ) striking the panels, classifying the micro-grid into one of four mutually exclusive operating environments: **ClearSkyState** (peak production), **PartialCloudState** (intermittent volatility), **OvercastState** (dense cloud cover), or **NighttimeState** (zero solar availability).

Simultaneously, the system assesses physical health by evaluating whether the `PhotovoltaicArray` `hasAnomaly`. It detects six distinct `AnomalyState` subclasses:

- **Clipping Anomaly:** Occurs when solar power saturates and hits the inverter’s maximum limits under intense irradiance.
- **Inverter Trip Anomaly:** A critical physical failure where power output drops to zero despite high available solar irradiance.
- **Ramp Anomaly:** Detected when the rate of change in power ( $\Delta P$ ) over a consecutive interval exceeds safe grid-frequency thresholds.
- **Soiling Anomaly:** A prolonged drop in conversion efficiency caused by physical panel surface contamination (dust, soil).
- **Overheating Anomaly:** High ambient temperatures trigger thermal derating, decreasing panel performance under strong sunlight.
- **Malfunction Anomaly:** Severe degradation where hardware output drops to a critically low, non-zero threshold without a clear environmental cause.

## Machine Learning Vulnerabilities: Aliasing Events

In parallel with physical faults, the system evaluates data-processing anomalies. If the chosen patch tokenization parameters conflict with the signal’s natural periodic cycles, the `PatchedSignal` logically `exhibitsAliasing`, triggering an `AliasingEvent`. These represent structural blind spots where information is permanently lost prior to AI forecasting:

- A `FrequencyBlindSpot` occurs when the signal’s fundamental frequency (or its secondary harmonic) matches the patch-induced frequency ( $f_p = f_s/S$ ), rendering cyclical trends invisible to the model (Hypotheses H1 and H1b).
- A `PhaseAliasingEffect` (Hypothesis H2) is flagged when a temporal phase offset misaligns with the rigid token boundaries ( $0 < \phi < S$ ), corrupting early representation features.
- An `OverlapEffect` (Hypothesis H3) identifies configuration gaps where a low-density overlap ratio ( $\gamma < 0.5$ ) introduces data redundancy or framing distortion.

## Closed-Loop Autonomous Action

To maintain grid equilibrium and mitigate faults, the `PhotovoltaicArray` `triggersAction` directives that map directly to an `AgentAction`. The autonomous `ControlAgent` is responsible for orchestrating the closed-loop feedback loop and `performs` four targeted interventions based on the inferred states:

- It schedules a `BatteryDispatchAction` to buffer power during `PartialCloudState` volatility or to substitute missing capacity during an `InverterTripAnomaly`.
- It executes a `CurtailGenerationAction` to throttle the inverter during `ClippingAnomaly` or `OverheatingAnomaly` events to safeguard structural integrity.
- It implements a `LoadAdjustmentAction` to dynamically down-shift internal consumer demand during `OvercastState` or `NighttimeState` deficits.
- It fires a `RampAlertAction` to warn the broader grid stability network the moment a dangerous `RampAnomaly` is diagnosed.

The core ontology architecture bridges the gap between cyber-physical energy systems and deep learning signal processing domain knowledge. It categorizes the application setting into five main conceptual pillars rooted under the ontological hierarchy:

- **Physical Subsystem (`pv:PhysicalComponent`):** Represents the hardware infrastructure of the micro-grid, including the solar array, electrochemical storage, consumer loads, and the macro grid tie.

- **Cognitive Layer (pv:ControlAgent):** The artificial intelligence system responsible for autonomous grid optimization, load balancing, dispatch scheduling, and proactive vulnerability mitigation.
- **Signal Processing Layer (pv:Signal, pv:PatchedSignal):** Tracks the raw telemetry inputs (both time-domain features and frequency-domain components) along with their tokenized representation configurations (patch dimensions, stride values) tailored for the Chronos foundation architecture.
- **Environmental Tracking (pv:WeatherObservation):** Captures concurrent meteorological snapshots (air temperature, wind vectors) influencing structural physics and electrical efficiency.
- **Inferred Operational States:** Dynamic metadata classified into generation levels (pv:GenerationState), systemic disruptions (pv:AnomalyState), control directives (pv:AgentAction), and signal degradation hazards (pv:AliasingEvent).

## Core Concept Taxonomy

The formal classes defined within the schema are explicitly detailed below:

- pv:PhotovoltaicArray** The primary renewable energy asset. Characterized by panel count and nameplate nominal capacity. Source of telemetry streams; associated with weather conditions and subject to environmental/electrical anomalies.
- pv:BatteryStorage** Local electrochemical reservoir capable of dynamic grid connection/isolation. Characterized by capacity and instantaneous State of Charge (SoC).
- pv:InternalLoad** Represents internal industrial or residential consumption demands. Subject to adjustable, demand-side management interventions.
- pv:NationalGrid** The external electrical macro-grid interface, acting as a bidirectional infinite bus for excess export or deficit imports.
- pv:Signal** Time-series measurement stream containing global horizontal irradiance ( $G_h$ ), tilted surface irradiance ( $G_{\text{tilt}}$ ), instantaneous power output ( $P_{\text{pv}}$ ), and analytical frequency descriptors (sampling rate, fundamental frequency, amplitude, phase).
- pv:PatchedSignal** The Chronos-specific tokenized structure derived from a raw signal. Governed by a selected window chunking mechanism (*patchSize*, *stride*), establishing a structural *overlapRatio* and a downstream *patchInducedFrequency*.

## Semantic Relations & Properties

### Object Properties (Entity-to-Entity Relationships)

The physical, functional, and inferred linkages across concepts are formalized using the following object properties:

- $\langle \text{pv:PhotovoltaicArray} \rangle \text{ pv:measures } \langle \text{pv:Signal} \rangle$
- $\langle \text{pv:PhotovoltaicArray} \rangle \text{ pv:hasWeatherObservation } \langle \text{pv:WeatherObservation} \rangle$
- $\langle \text{pv:PhotovoltaicArray} \rangle \text{ pv:hasGenerationState } \langle \text{pv:GenerationState} \rangle$  (*Inferred via SWRL*)
- $\langle \text{pv:PhotovoltaicArray} \rangle \text{ pv:hasAnomaly } \langle \text{pv:AnomalyState} \rangle$  (*Inferred via SWRL*)
- $\langle \text{pv:PhotovoltaicArray} \rangle \text{ pv:triggersAction } \langle \text{pv:AgentAction} \rangle$  (*Inferred via SWRL*)
- $\langle \text{pv:Signal} \rangle \text{ pv:tokenizes } \langle \text{pv:PatchedSignal} \rangle$
- $\langle \text{pv:PatchedSignal} \rangle \text{ pv:exhibitsAliasing } \langle \text{pv:AliasingEvent} \rangle$  (*Inferred via SWRL*)
- $\langle \text{pv:BatteryStorage} \rangle \text{ pv:buffers } \langle \text{pv:PhotovoltaicArray} \rangle$
- $\langle \text{pv:PhysicalComponent} \rangle \text{ pv:connectedTo } \langle \text{pv:PhysicalComponent} \rangle$  (*Symmetric Property*)



## Data Properties (Entity Attributes)

Key quantitative attributes defining system conditions include:

- **Electrical/Solar:**  $\text{pv:pvPowerW}$  ( $P_{\text{pv}}$ ),  $\text{pv:nominalPowerW}$  ( $P_{\text{nom}}$ ),  $\text{pv:tiltedIrradianceWm2}$  ( $G_{\text{tilt}}$ ),  $\text{pv:deltaPowerW}$  ( $\Delta P$ ).
- **Meteorological:**  $\text{pv:airTemperatureC}$  ( $T_{\text{air}}$ ),  $\text{pv:windSpeedMs}$  ( $W_s$ ).
- **Tokenization Architecture:**  $\text{pv:patchSize}$  ( $N_p$ ),  $\text{pv:stride}$  ( $S$ ),  $\text{pv:overlapRatio}$  ( $\gamma$ ),  $\text{pv:patchInducedFrequency}$  ( $f_p$ ).
- **Signal Descriptors:**  $\text{pv:samplingFrequencyHz}$  ( $f_s$ ),  $\text{pv:fundamentalFrequencyHz}$  ( $f_0$ ),  $\text{pv:phaseRad}$  ( $\phi$ ).

## Knowledge-Driven Inference System (SWRL Rules)

Real-time diagnostic, classification, and mitigation logic are automated using a rule-based engine. The standard logical form of these production rules is detailed below:

### Environmental Context Extraction

$$\begin{aligned} \text{PhotovoltaicArray}(?pv) \wedge \text{measures}(?pv, ?s) \wedge \text{tiltedIrradianceWm2}(?s, ?g) \wedge (?g \geq 800.0) \\ \longrightarrow \text{hasGenerationState}(?pv, \text{ClearSky}) \end{aligned} \quad (24)$$

$$\begin{aligned} \text{PhotovoltaicArray}(?pv) \wedge \text{measures}(?pv, ?s) \wedge \text{tiltedIrradianceWm2}(?s, ?g) \\ \wedge (?g \geq 150.0) \wedge (?g < 800.0) \longrightarrow \text{hasGenerationState}(?pv, \text{PartialCloud}) \end{aligned} \quad (25)$$

$$\begin{aligned} \text{PhotovoltaicArray}(?pv) \wedge \text{measures}(?pv, ?s) \wedge \text{tiltedIrradianceWm2}(?s, ?g) \\ \wedge (?g > 0.0) \wedge (?g < 150.0) \longrightarrow \text{hasGenerationState}(?pv, \text{Overcast}) \end{aligned} \quad (26)$$

$$\begin{aligned} \text{PhotovoltaicArray}(?pv) \wedge \text{measures}(?pv, ?s) \wedge \text{tiltedIrradianceWm2}(?s, ?g) \wedge (?g \leq 0.0) \\ \longrightarrow \text{hasGenerationState}(?pv, \text{Nighttime}) \end{aligned} \quad (27)$$

### Cyber-Physical Anomaly Diagnostics

- **Inverter Clipping Anomaly (R5):**

$$\begin{aligned} \text{PhotovoltaicArray}(?pv) \wedge \text{measures}(?pv, ?s) \wedge \text{pvPowerW}(?s, ?p) \wedge \text{nominalPowerW}(?pv, ?nom) \\ \wedge (?p \geq ?nom) \wedge \text{tiltedIrradianceWm2}(?s, ?g) \wedge (?g > 800.0) \\ \longrightarrow \text{hasAnomaly}(?pv, \text{Clipping}) \end{aligned} \quad (28)$$

- **Inverter Trip Anomaly (R6):**

$$\begin{aligned} \text{PhotovoltaicArray}(?pv) \wedge \text{measures}(?pv, ?s) \wedge \text{pvPowerW}(?s, ?p) \wedge (?p \leq 0.0) \\ \wedge \text{tiltedIrradianceWm2}(?s, ?g) \wedge (?g > 200.0) \longrightarrow \text{hasAnomaly}(?pv, \text{InverterTrip}) \end{aligned} \quad (29)$$

- **Ramp Event Anomaly (R7):**

$$\text{PhotovoltaicArray}(?pv) \wedge \text{measures}(?pv, ?s) \wedge \text{deltaPowerW}(?s, ?dp) \wedge (?dp > 500.0) \longrightarrow \text{hasAnomaly}(?pv, \text{Ramp}) \quad (30)$$

- **Soiling Performance Deficit Anomaly (R8):**

$$\begin{aligned} & \text{PhotovoltaicArray}(?pv) \wedge \text{measures}(?pv, ?s) \wedge \text{tiltedIrradianceWm2}(?s, ?g) \wedge (?g > 400.0) \\ & \wedge \text{pvPowerW}(?s, ?p) \wedge (?p > 100.0) \wedge \text{swrlb:divide}(?r, ?p, ?g) \wedge (?r < 0.3) \\ & \longrightarrow \text{hasAnomaly}(?pv, \text{Soiling}) \end{aligned} \quad (31)$$

- **Thermal Overheating Derating Anomaly (R9):**

$$\begin{aligned} & \text{PhotovoltaicArray}(?pv) \wedge \text{hasWeatherObservation}(?pv, ?w) \wedge \text{airTemperatureC}(?w, ?t) \wedge (?t > 35.0) \\ & \wedge \text{measures}(?pv, ?s) \wedge \text{tiltedIrradianceWm2}(?s, ?g) \wedge (?g > 600.0) \\ & \wedge \text{pvPowerW}(?s, ?p) \wedge \text{swrlb:divide}(?r, ?p, ?g) \wedge (?r < 0.4) \\ & \longrightarrow \text{hasAnomaly}(?pv, \text{Overheating}) \end{aligned} \quad (32)$$

## Autonomous Closed-Loop Control Actuation

Based on inferred operating profiles, actions are dynamically mapped to handle grid stresses:

$$\text{pv:hasAnomaly}(?pv, \text{pv:Ramp}) \longrightarrow \text{pv:triggersAction}(?pv, \text{pv:RampAlert}) \quad (33)$$

$$\text{pv:hasAnomaly}(?pv, \text{pv:Clipping}) \vee \text{pv:hasAnomaly}(?pv, \text{pv:Overheating}) \quad (34)$$

$$\longrightarrow \text{pv:triggersAction}(?pv, \text{pv:CurtailGeneration}) \quad (35)$$

$$\text{pv:hasAnomaly}(?pv, \text{pv:InverterTrip}) \vee \text{pv:hasGenerationState}(?pv, \text{pv:PartialCloud}) \quad (36)$$

$$\longrightarrow \text{pv:triggersAction}(?pv, \text{pv:BatteryDispatch}) \quad (37)$$

$$\text{pv:hasGenerationState}(?pv, \text{pv:Overcast}) \vee \text{pv:hasGenerationState}(?pv, \text{pv:Nighttime}) \quad (38)$$

$$\longrightarrow \text{pv:triggersAction}(?pv, \text{pv:LoadAdjustment}) \quad (39)$$

## Chronos Structural Aliasing Hypotheses

The key scientific contribution of this architecture lies in identifying information loss bounds during time-series patching. The model detects structural blind spots via frequency domain reasoning:

### Hypothesis H1 & H1b: Patch-Induced Frequency Blind Spots

The patch tokenization process introduces an artificial operating frequency profile  $f_p$ , mathematically defined by the relationship:

$$f_p = \frac{f_s}{S} \quad (40)$$

Where  $f_s$  represents the raw Nyquist sampling rate and  $S$  denotes the token step stride.

- **Primary Blind Spot (H1):** If the continuous signal's structural fundamental frequency ( $f_0$ , such as the daily diurnal solar cycle) aligns directly with the token boundaries, a tracking blind spot forms at

the fundamental harmonic region ( $n = 1$ ):

$$\begin{aligned} & \text{Signal}(?s) \wedge \text{tokenizes}(?s, ?ps) \wedge \text{fundamentalFrequencyHz}(?s, ?f_0) \wedge \text{patchInducedFrequencyHz}(?ps, ?f_p) \\ & \wedge \text{swrlb:divide}(?r, ?f_0, ?f_p) \wedge (?r \geq 0.9) \wedge (?r \leq 1.1) \\ & \longrightarrow \text{exhibitsAliasing}(?ps, \text{H1\_BlindSpot}) \end{aligned} \quad (41)$$

- **Secondary Harmonic Blind Spot (H1b):** Re-occurs at the second harmonic ( $n = 2$ ) boundary layer:

$$\begin{aligned} & \text{Signal}(?s) \wedge \text{tokenizes}(?s, ?ps) \wedge \text{fundamentalFrequencyHz}(?s, ?f_0) \wedge \text{patchInducedFrequencyHz}(?ps, ?f_p) \\ & \wedge \text{swrlb:divide}(?r, ?f_0, ?f_p) \wedge (?r \geq 1.85) \wedge (?r \leq 2.15) \\ & \longrightarrow \text{exhibitsAliasing}(?ps, \text{H1b\_BlindSpot}) \end{aligned} \quad (42)$$

## Hypothesis H2: Phase Aliasing Effect

Vulnerabilities appear when temporal offsets misalign with chunk window anchors, corrupting early representation extraction features ( $0 < \phi < S$ ):

$$\begin{aligned} & \text{Signal}(?s) \wedge \text{tokenizes}(?s, ?ps) \wedge \text{phaseRad}(?s, ?\phi) \wedge \text{stride}(?ps, ?S) \\ & \wedge (?\phi > 0.0) \wedge (? \phi < ?S) \longrightarrow \text{exhibitsAliasing}(?ps, \text{H2\_Phase}) \end{aligned} \quad (43)$$

## Hypothesis H3: Overlap Information Distortion

When the overlap configuration ratio ( $\gamma = \frac{N_p - S}{N_p}$ ) drops below a critical information-theoretic bound ( $\gamma < 0.5$ ), consecutive contextual patches display heavy text-token gaps, rendering trend forecasts highly unstable:

$$\text{PatchedSignal}(?ps) \wedge \text{overlapRatio}(?ps, ?or) \wedge (?or < 0.5) \longrightarrow \text{exhibitsAliasing}(?ps, \text{H3\_Overlap}) \quad (44)$$

# Structural Aliasing Expected Impact

## Formalization of Falsifiable Hypotheses

To ground the experimental validation within the semantic framework established by the system ontology, we map the digital signal processing phenomena directly to the ontological layer. Let a continuous-time physical signal (such as `pv:pvPowerW` or `pv:tiltedIrradianceWm2`) be sampled uniformly at a rate  $f_s$  (`pv:samplingFrequencyHz`) and transformed into a tokenized sequence via a `pv:PatchedSignal` operator defined by patch size  $P$  (`pv:patchSize`) and translation stride  $S$  (`pv:stride`).

The architectural grid frequency  $f_p$  (`pv:patchInducedFrequencyHz`) governs the macro-temporal resolution of the token space and is defined as:

$$f_p = \frac{f_s}{S} \quad (45)$$

Based on this parametric environment, we define three distinct, falsifiable hypotheses to characterize the boundaries of structural aliasing.

### Hypothesis 1 ( $H_1$ ): Frequency-Dependent Blind Spots

- **Ontological Alignment:** `pv:FrequencyBlindSpot` (asserted via SWRL inference rules R18 and R19).

- **Statement:** When the fundamental frequency  $f$  (`pv:fundamentalFrequencyHz`) of an input time-series signal matches an integer multiple of the patch-induced frequency  $f_p$ , the tokenization layer maps distinct cyclical variations into static, identical latent representations, creating discrete informational blind spots across the spectrum.
- **Mathematical Boundary:** An aliasing event is triggered when the frequency ratio satisfies the harmonic condition:

$$f = n \cdot f_p = n \left( \frac{f_s}{S} \right), \quad n \in \mathbb{N}^+ \quad (46)$$

Under this boundary condition, the column rank of the local trajectory matrix collapses to unity, rendering the internal dynamics of the patch unobservable to downstream self-attention mechanisms.

- **Falsifiability Criterion:** This hypothesis is falsified if empirical probing models or closed-loop reconstruction tasks demonstrate uniform representation fidelity across the operational band  $[2, \frac{f_s}{2}]$  Hz, without exhibiting statistically significant localized performance degradation or info-loss anomalies at the exact integer harmonics of  $f_p$ .

### Hypothesis 2 ( $H_2$ ): Phase Alignment and Temporal Displacement

- **Ontological Alignment:** `pv:PhaseAliasingEffect` (asserted via SWRL inference rule R20).
- **Statement:** The magnitude of structural information loss at an aliased harmonic frequency is non-deterministic and varies as a function of the temporal alignment between the signal’s peak-trough dynamics and the rigid boundaries of the tokenization grid.
- **Mathematical Boundary:** For a fixed harmonic frequency  $f \in \mathcal{F}_{\text{lock}}$ , translating the initial phase offset  $\phi$  (`pv:phaseRad`) within the discrete-time sinusoid:

$$x[n] = \sin \left( 2\pi \frac{f}{f_s} n + \phi \right), \quad \phi \in [0, 2\pi) \quad (47)$$

modifies the specific sub-sequence frozen inside each individual patch window. If the phase shift satisfies  $\phi \in (0, S)$ , it induces a structural misalignment that alters the component-wise values of consecutive token vectors, systematically modulating the token space fidelity.

- **Falsifiability Criterion:** This hypothesis is falsified if information-theoretic tracking metrics (such as the Space Saving metric  $SV$ ) or empirical probing classification accuracies remain strictly invariant across all randomly generated phase displacements  $\phi \sim \text{Uniform}(0, 2\pi)$  at a locked harmonic frequency, proving that boundary location exercises no control over latent feature degradation.

### Hypothesis 3 ( $H_3$ ): Geometric Configuration and Overlap Mechanics

- **Ontological Alignment:** `pv:OverlapEffect` (asserted via SWRL inference rule R21).
- **Statement:** Altering the geometric relationship between the patch window length and the translational step size shifts the failure mode of the token space, moving the representation from absolute vector degeneracy to high-dimensional contextual redundancy.
- **Mathematical Boundary:** The geometric interaction is dictated by the token overlap ratio  $or$  (`pv:overlapRatio`):

$$or = \frac{P - S}{P} \quad (48)$$

When  $or = 0$  ( $P = S$ ), the windowing is strictly contiguous and non-overlapping; satisfying Equation (2) forces consecutive tokens to be mathematically identical ( $\mathbf{t}_j = \mathbf{t}_{j+1}$ ), wiping out temporal gradients. When  $or \rightarrow 1$  ( $S \ll P$ ), successive patches share massive historical context; although the stride-bound phase-locking condition  $x[n + S] = x[n]$  still holds, the expanded patch dimension  $P$  forces tokens to capture intra-patch phase variations that break the absolute vector identity, shifting the error into an over-smoothed informational redundancy.

- **Falsifiability Criterion:** This hypothesis is falsified if varying the architectural parameters  $P$  and  $S$  while keeping a locked signal frequency  $f$  yields an identical trajectory matrix rank profile and an identical distribution of downstream transformer attention weights, proving that the geometric layout of the patch grid does not modulate the structural aliasing expression.

## Signal Generation

Simple sinusoid generation. TSMixup and KernelSynth description, algorithm pseudo-code.